Amazon Web Services: CloudWatch

Topics

- What is CloudWatch?
- CloudWatch Features
- Types of Metrics
- Demo
- Application Monitoring with CloudWatch Logs

What is CloudWatch?

- CloudWatch is a monitoring service for cloud resources and the applications that run in AWS

- Used to gain system-wide visibility into resource utilization, application performance, and operational health

Amazon CloudWatch

https://aws.amazon.com/cloudwatch/faqs/

CloudWatch Features

System Metrics	Application Metrics	Alarms
• These are metrics that you get out of the box • The more managed the service is, the more you get	• By installing the CloudWatch Agent, you can get information from inside your resources	• What is the point of collecting data if you don't do something with it? • Alarms can alert you when something goes wrong

Types of Metrics

- Default – provided out of the box and do not require any additional configurations
- Custom – provided by the CloudWatch agent installed on a host

	Default	Custom
CPU Utilization	✓	
Network Throughput	✓	
EC2 Memory Utilizations		✓
EBS Storage Capacity		✓

Demo: CloudWatch

Application Monitoring with CloudWatch Logs

What Is CloudWatch Logs?

- A Tool to monitor, store, and access log files from a variety of different sources
- Provides the ability to query the logs and look for potential issues or data that is relevant

3 CloudWatch Logs Terms

- **Log Event** – this is a record of what happened; contains a timestamp and the data

- **Log Stream** – a collection of log events from the same source create a log stream. Think of one continuous set of logs from a single instance

- **Log Group** – a collection of log streams. For example, you'd group all your Apache web server logs across hosts together.

CloudWatch Logs Features

Filter Patterns	CloudWatch Logs Insights	Alarms
• Look for specific terms in your logs • Example: 400 errors in your webserver logs	• Allows you to use SQL-like interactive solution to query your logs	• Once you have identified your trends or patterns, it's time to alert on them

Checkpoint

Which of the following can be viewed using the default CloudWatch metrics?

Choose 2

1. EBS storage capacity
2. Network throughput
3. EC2 memory utilization
4. CPU utilization

Which web service and feature can be used to monitor, store, and access your log files from Amazon EC2 instances, AWS CloudTrail, Route 53, and other sources?

1. CloudWatch
2. CloudWatch Logs
3. Amazon EventBridge (formerly called Amazon CloudWatch Events)
4. Log Events

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